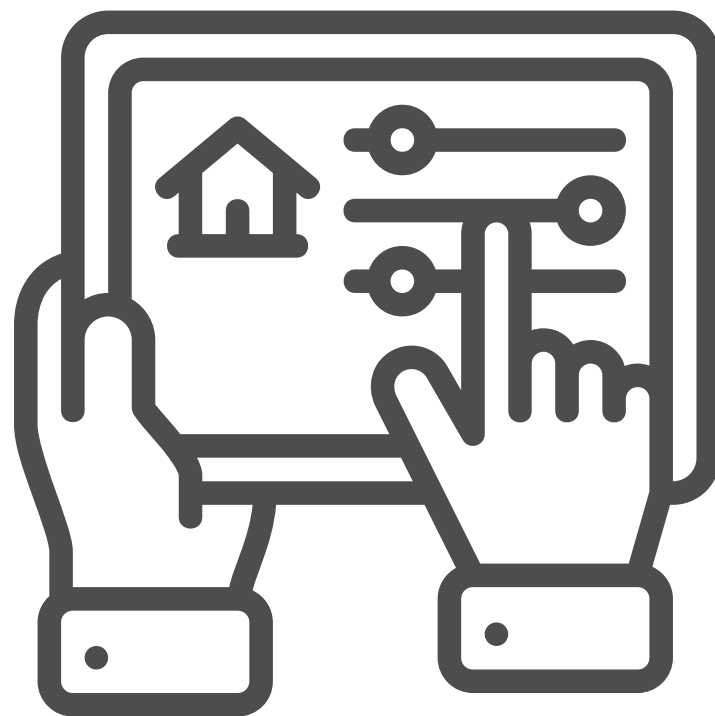




SMART GAS LEAKAGE DETECTION & AUTOMATIC SAFETY SYSTEM

BCA | CS&F | SEC. BCACS31 | GROUP NO. 8



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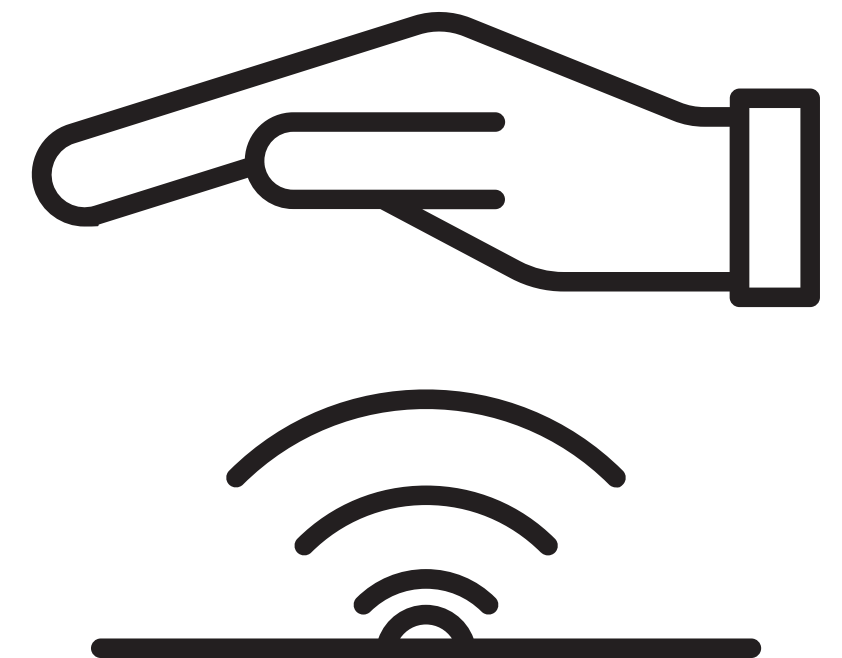
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1. Project Overview

- An Arduino-based safety prototype for early detection of LPG/flammable gas leakage.
- MQ-2 continuously senses gas concentration and sends an analog signal to Arduino UNO.
- When a threshold is crossed, the system activates a buzzer, exhaust fan and servo-based safety mechanism.
- SIM800L GSM module sends an emergency SMS to a predefined mobile number.
- The project combines IoT concepts, embedded systems, automation and safety engineering.



2. Problem Statement & Motivation

The Problem

Gas leakage can remain unnoticed, especially when occupants are away or asleep.

The Risk

Accumulated LPG/flammable gas may create fire, explosion and health hazards.

The Need

A low-cost system should detect leakage and trigger multiple safety actions quickly.

- Reduce response time between leakage detection and safety action.
- Provide both local warning and remote notification.
- Demonstrate a practical smart-home/IoT safety concept.

3. Objectives

- Detect LPG/flammable gas leakage using an MQ-2 sensor.
- Process sensor readings using Arduino UNO.
- Automatically activate an audible warning using a buzzer.
- Turn ON an exhaust fan through a relay to improve ventilation.
- Operate a servo-based prototype mechanism for regulator/valve control.
- Send an emergency SMS through SIM800L GSM.
- Build a low-cost, demonstrable academic safety prototype.

4. Hardware & Software Components

MQ-2 Gas Sensor

Detects LPG/smoke/flammable gases.

Arduino UNO

Main controller and decision-making unit.

SIM800L GSM

Sends emergency SMS alerts.

Relay Module

Controls the exhaust fan/load.

Exhaust Fan

Provides ventilation when leakage is detected.

Servo Motor

Moves the prototype regulator/valve mechanism.

Buzzer

Provides local audible warning.

Arduino IDE

Used to write, compile and upload firmware.

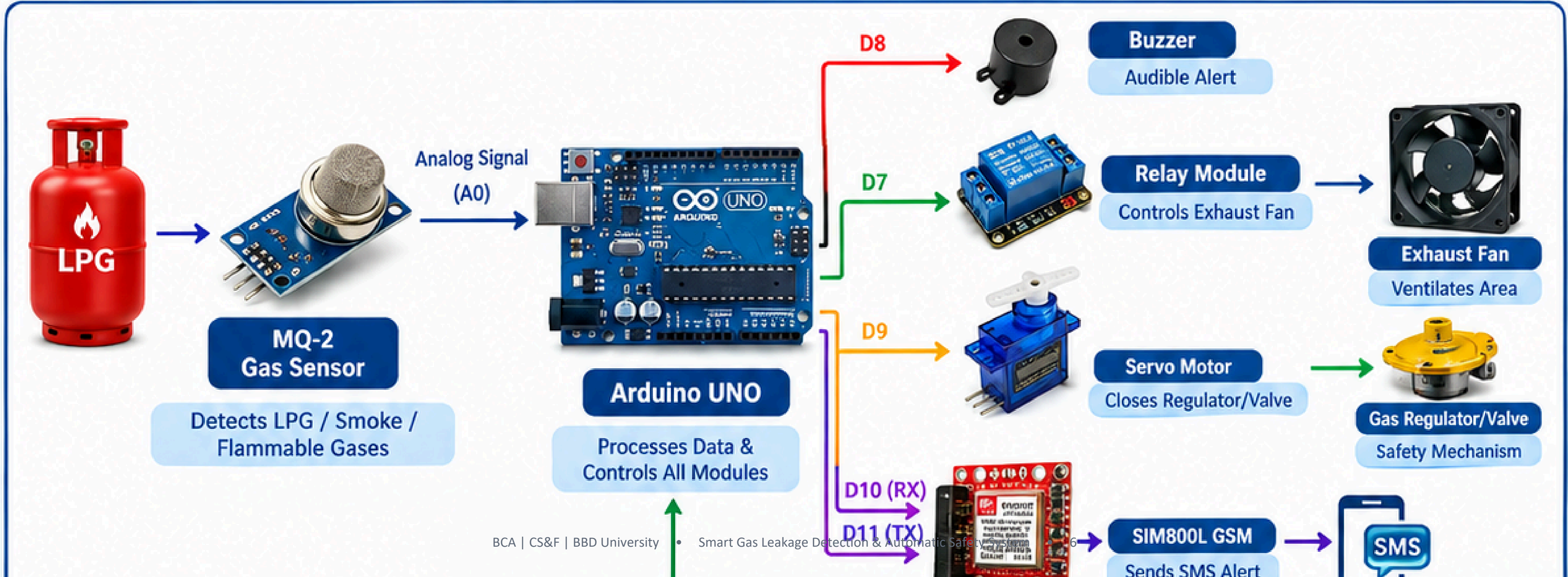
5. Complete Working Flow



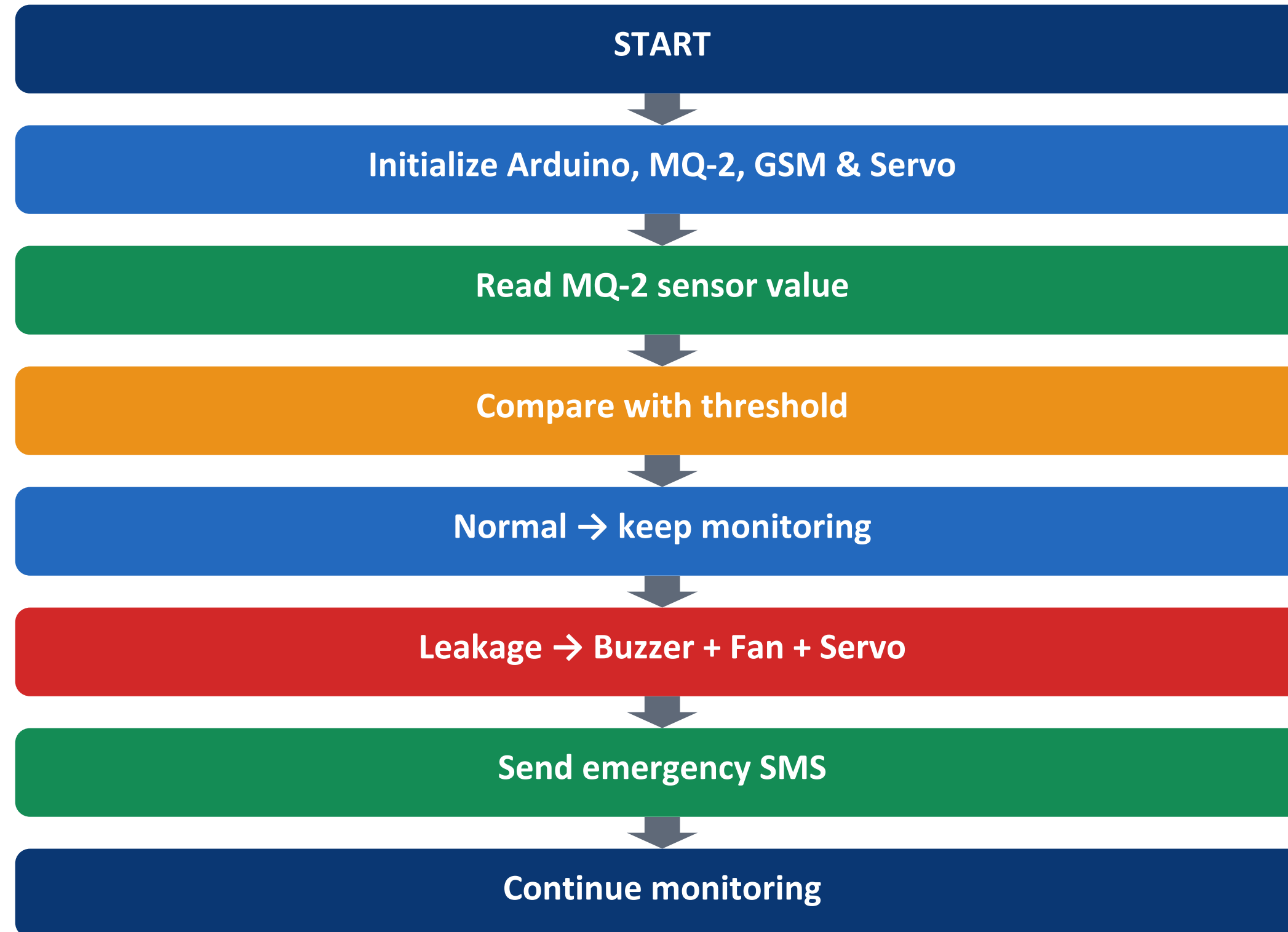
SMART GAS LEAKAGE DETECTION & AUTOMATIC SAFETY SYSTEM

Detect Early
Act Fast
Stay Safe

WORKING DIAGRAM



6. System Logic / Algorithm



7. Arduino UNO Pin Configuration

Component / Signal	Arduino Pin	Purpose
MQ-2 AO	A0	Analog gas reading
Relay IN	D7	Exhaust fan control
Buzzer +	D8	Audible alert
Servo Signal	D9	Valve/regulator mechanism
SIM800L TX	D10	Arduino RX (SoftwareSerial)
SIM800L RX	D11	Arduino TX (SoftwareSerial)
Common GND	GND	Shared reference

8. GSM SMS Alert using SIM800L

1. Detect

Arduino identifies a gas value above the configured threshold.

2. Trigger

SoftwareSerial communicates with SIM800L using GSM AT commands.

3. Notify

SIM800L sends an emergency SMS to the configured phone number.

Example alert:

“WARNING! GAS LEAKAGE DETECTED. Please check the gas regulator/cylinder. Automatic safety system activated.”

9. Automatic Safety Actions

Buzzer

Provides an immediate local warning when the gas threshold is crossed.

Relay + Fan

Relay switches the exhaust fan ON to help ventilate the affected area.

Servo Mechanism

Servo moves the prototype regulator/valve mechanism toward the safety position.

For a real LPG installation, use a certified automatic shut-off valve and proper electrical isolation.

10. Software & Code Architecture

- Language: Embedded C/C++ for Arduino.
- IDE: Arduino IDE for writing, compiling and uploading firmware.
- Servo.h controls the servo motor position.
- SoftwareSerial.h provides serial communication with SIM800L.
- MQ-2 analog value is compared against GAS_THRESHOLD.
- Conditional logic activates the safety devices only when the threshold is crossed.
- SMS is sent once per leakage event to avoid repeated alerts.

11. Testing & Expected Results

Test Case	Condition	Expected Result	Status
Normal Air	MQ-2 below threshold	No alarm; continue monitoring	✓
Gas Level Rising	Sensor value increases	Arduino detects abnormal condition	✓
Threshold Crossed	Gas value $\geq$ threshold	Buzzer + fan + servo activated	✓
SMS Test	GSM network available	Emergency SMS delivered	✓
Recovery	Gas level returns safe	System returns to monitoring	✓

12. Advantages

- Low-cost and easy-to-understand prototype.
- Real-time detection with automatic response.
- Multiple layers of safety: alarm, ventilation, valve mechanism and SMS.
- Remote alert capability through GSM.
- Modular design allows future IoT/cloud integration.
- Useful as an academic demonstration of embedded automation and smart safety.

13. Limitations & Safety Considerations

- MQ-2 is a general-purpose sensor; it requires calibration and can respond to multiple gases/smoke.
- Threshold value must be tuned using actual sensor readings and the intended environment.
- SIM800L needs a suitable power supply and GSM network coverage.
- Servo-based regulator control shown here is a prototype mechanism, not a certified gas shut-off device.
- AC fan/load switching requires an appropriately rated relay, enclosure and safe isolation.
- Never release LPG uncontrolled for testing; use safe, supervised testing methods.

14. Future Scope

- IoT dashboard for live gas-level monitoring.
- Mobile application and push notifications.
- Cloud database and event logging.
- Secure authentication and encrypted IoT communication.
- Battery backup for power failure situations.
- AI/ML-based anomaly detection and sensor trend analysis.
- Integration with certified automatic gas shut-off hardware.
- Cyber-security hardening for connected devices and APIs.

15. Conclusion

- The project demonstrates an automated approach to gas-leakage safety using low-cost embedded components.
- MQ-2 provides sensing, Arduino UNO makes the decision, and the connected actuators provide immediate local safety actions.
- SIM800L adds remote SMS notification, improving awareness when the user is not beside the system.
- The prototype provides a strong foundation for future IoT, cloud and cyber-security enhancements.

THANK YOU

16. Team & Submission

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